GestureLab CAD

A 3D engineering tutor you can talk to, point at, and learn from, in your own language

SOLUTION PITCH
Great Indian AI Internship Challenge 2026 by Gnani AI
Jugaad Genius Award entry

A student points at a part of a 3D machine, asks a question out loud in English, Telugu or Hindi, and the machine answers: the right parts light up and move while Gnani Timbre explains, in the same language the student spoke.

The problem

Engineering students can **see** a machine, but they cannot **ask** it anything. Textbooks are flat, 3D viewers show shape but explain nothing, cut-away models are rare and costly, and doubts come late at night when no teacher is around. Chatbots write paragraphs but cannot see the part the student means. And for millions of students who think in Telugu or Hindi, almost every answer arrives in English.

Our solution: the machine becomes the teacher

1 Point	2 Ask	3 Gnani Prisma	4 AI lesson	5 Gnani Timbre
at a part with a finger (webcam) or the mouse	out loud; recording stops when you finish	turns speech into text and detects the language	from the model's real parts data, every step validated	speaks each step as its parts light up and move

No language menu: every question is answered in the language it was asked, switching English, Telugu and Hindi turn by turn. The tutor then checks understanding with a two-question spoken quiz, and can be interrupted at any time.

Why it is creative use of Gnani AI (not just another chatbot)

Gnani Prisma = the controls	A spoken question becomes a validated action on a real 3D machine. Automatic Indic language detection; the model's part names are sent as key terms.
Gnani Timbre = synced with motion	Each sentence is spoken exactly when its parts light up or the assembly opens, so hearing and seeing happen together, in the student's language.
Voice + hands together	Point at a part and ask about “this”. Hand tracking runs in the browser: video never leaves the device.
Grounded and honest	Answers come from the model's real parts data; every action is checked before it runs; reasoning is labelled, never shown as simulation.

Key features

- 4 interactive 3D models: planetary gearbox, steam engine, RC building frame, human eye
- Exploded view, X-ray view, parts inspector and relationship map
- Hand gestures: orbit, zoom, point-to-select, two-hand explode
- Multilingual voice tutor with quiz, interruption and clear error messages
- Answer starts speaking within seconds (first sentence ~4 s in live tests)

Tech stack

Frontend: React 19, TypeScript, Vite, Three.js, Web Audio
Hand tracking: MediaPipe Hand Landmarker (pretrained, in-browser)
Backend: Python, FastAPI (keeps all API keys on the server)
AI: Gnani Prisma (STT), Gnani Timbre (TTS), Gemini (lesson plans)
Hosting: Vercel + Render · 983 frontend + 222 backend tests

Who benefits

- Mechanical, civil and diploma students meeting a machine for the first time
- Polytechnic and ITI learners without cut-away models or labs
- Students who learn best in Telugu or Hindi
- Teachers demonstrating to a class with a laptop and projector

Feasible and scalable

Runs in any browser with a laptop, webcam and microphone. A new subject is just a new 3D model with its parts data; the tutor, voice and gestures work unchanged. More Indian languages are the natural next step.

What's next: from 4 machines to any machine

Dynamic models	Say it, see it	Live physics	Every Indian language
A teacher uploads any 3D/CAD file; AI names the parts and maps how they connect, so a new machine becomes a talking tutor in minutes.	“Show me a car differential”: the tutor finds and loads the right model by voice and starts teaching it.	Watch torque flow and loads move through the parts, and see what really happens when a gear is removed.	Grow from English, Telugu and Hindi to Tamil, Kannada, Marathi, Bengali and more with Gnani's voices.

Try it live: https://3dexplodeview-frontend.vercel.app/

Source code: https://github.com/munirajabommisetty-cmyk/3D_EXPLODE_VIEW

Status: working, deployed prototype. Not yet evaluated in a classroom study. **Every student deserves to ask why. GestureLab CAD lets them ask the machine itself.**